INTEGRAL BALANCES & REYNOLDS TRANSPORT

Topic 3: Conservation Laws (Integral Form)

Practical Subchapters & Problem-Solving Guide for 2nd Year Aerospace Engineering

Theory Syllabus | Problems & Exams

Modular Practical Subchapters

Select a subchapter for in-depth theory, step-by-step algorithms, and canonical exam walkthroughs

SUBCHAPTER 3.1 — Kinematic Tool

Reynolds Transport Theorem & Control Volumes

Control volume selection strategy (fixed, moving, deformable), the moving reference frame trick for unsteady waves, and boundary decomposition.

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt}\int_{V_c} \rho b\, dV + \int_{\Sigma_c} \rho b[(v - v_c)\cdot n]\, d\sigma$$

Canonical: The Hydraulic Bore / Moving Shock (CL2) →

SUBCHAPTER 3.2 — Continuity

Conservation of Mass in Problem Solving

Integral continuity balances, incompressible mass flow conservation, 1D streamtube approximations, and unsteady storage in draining tanks.

$$\frac{d}{dt}\int_{V_c} \rho\, dV + \int_{\Sigma_c} \rho(v\cdot n)\, d\sigma = 0$$

Canonical: Airfoil Wake Deficit Mass Balance (CL1) →

SUBCHAPTER 3.3 — Reaction Forces

Linear Momentum & Reaction Forces

Breakdown of the 5 momentum terms, the atmospheric gauge pressure trick, fluid vs solid force conventions, and the 5-stage exam algorithm.

$$\sum F_{\text{ext}} = \frac{d}{dt}\int_{V_c} \rho v\, dV + \int_{\Sigma_c} \rho v[(v - v_c)\cdot n]\, d\sigma$$

Canonical: Pipe Elbow (CL3), Rocket Thrust (CL5), Airfoil Drag (CL1) →

SUBCHAPTER 3.4 — Turbomachinery

Angular Momentum & Turbomachinery Torque

Strategic pivot selection to eliminate unknown reactions, moment balance of 1D streamtubes, and Euler's turbomachinery equation for rotors.

$$\sum M_{x_0} = \sum \dot{m}_{\text{out}}(r_{\text{out}} \times v_{\text{out}}) - \sum \dot{m}_{\text{in}}(r_{\text{in}} \times v_{\text{in}})$$

Canonical: Pipe Bend Torque (CL3/CL15) & Euler Rotor →

SUBCHAPTER 3.5 — 1st Law

Total Energy Conservation & Thermodynamics

Enthalpy grouping $h = e + p/\rho$, stagnation enthalpy h_0, engineering energy balance with heat transfer, and mechanical shaft work.

SUBCHAPTER 3.6 — Navier-Poisson

Stress Tensor, Navier–Poisson & Skin Friction

Cauchy stress decomposition into hydrostatic pressure and viscous deviator $\bar{\bar{\tau}}'$, wall shear stress τ_w, and skin friction drag integration.

$$\dot{Q} + \dot{W}_{\text{shaft}} = \sum \dot{m}_{\text{out}} h_{0,\text{out}} - \sum \dot{m}_{\text{in}} h_{0,\text{i}}$$

Canonical: Mixing Stream Temperature (CL15) & Fan Power (CL10) →

$$\bar{\bar{\tau}} = -p\bar{\bar{I}} + 2\mu\bar{\bar{T}}_d + \left(\mu_B - \frac{2}{3}\mu\right)(\nabla \cdot v)\bar{\bar{I}}$$

Canonical: Wall Friction Drag & Rotating Viscometer Torque →

Master Formula Cheat Sheet (Eqs. 3.1 to 3.46)

`Complete reference handbook from official UC3M lecture notes`

Eq. #	Mathematical Formulation
3.1	$$\Phi(t) = \int_{V_f(t)} \phi(x,t)\, dV$$
3.2	$$\frac{d}{dt}\left[\int_{V_f(t)} \phi dV\right] = \lim_{\Delta t \to 0} \frac{1}{\Delta t}\left[\int_{V_f(t+\Delta t)} \phi(t+\Delta t) dV - \int_{V_f(t)} \phi(t) dV\right]$$
3.3	$$\lim_{\Delta t \to 0} \frac{1}{\Delta t}\left[\int_{V_f(t)} [\phi(t+\Delta t) - \phi(t)] dV + \int_{V_f(t+\Delta t) - V_f(t)} \phi(t+\Delta t) dV\right]$$
3.4	$$\lim_{\Delta t \to 0} \frac{1}{\Delta t} \int_{V_f(t)} [\phi(t+\Delta t) - \phi(t)] dV = \int_{V_f(t)} \frac{\partial \phi}{\partial t} dV$$
3.5	$$\lim_{\Delta t \to 0} \frac{1}{\Delta t} \int_{V_f(t+\Delta t) - V_f(t)} \phi(t+\Delta t) dV = \int_{\Sigma_f(t)} \phi v \cdot n \, d\sigma$$
3.6	$$\frac{d}{dt}\left[\int_{V_f(t)} \phi \, dV\right] = \int_{V_f(t)} \frac{\partial \phi}{\partial t} \, dV + \int_{\Sigma_f(t)} \phi \, v \cdot n \, d\sigma$$
3.7	$$\frac{d}{dt}\left[\int_{V_c(t)} \phi \, dV\right] = \int_{V_c(t)} \frac{\partial \phi}{\partial t} \, dV + \int_{\Sigma_c(t)} \phi \, v_c \cdot n \, d\sigma$$
3.8	$$\frac{d}{dt}\left[\int_{V_f(t)} \phi \, dV\right] = \frac{d}{dt}\left[\int_{V_c(t)} \phi \, dV\right] + \int_{\Sigma_c(t)} \phi \, (v - v_c) \cdot n \, d\sigma$$
3.9	$$\frac{d}{dt}\left[\int_{V_f(t)} \rho \, dV\right] = 0$$
3.10	$$\frac{d}{dt}\left[\int_{V_c(t)} \rho \, dV\right] + \int_{\Sigma_c(t)} \rho (v - v_c) \cdot n \, d\sigma = 0$$

Eq. #	Mathematical Formulation	
3.11	$$\int_{V_0} \frac{\partial \rho}{\partial t}\, dV + \int_{\Sigma_0} \rho v \cdot n\, d\sigma = 0$$	
3.12	$$dF_m = \rho f_m(x,t)\, dV$$	
3.13	$$f_m = g - a_0 - \frac{d\Omega}{dt} \wedge x - \Omega \wedge (\Omega \wedge x) - 2\Omega \wedge v$$	
3.14	$$g - a_0 - \Omega \wedge (\Omega \wedge x) = -\nabla U$$	
3.15	$$dF_s = f_n(n,x,t)\, d\sigma$$	
3.16	$$dA f_n - dA_1 f_1 - dA_2 f_2 - dA_3 f_3 = 0$$	
3.17	$$f_n = \bar{\bar{\tau}} \cdot n$$	
3.18	$$\bar{\bar{\tau}} = [\tau_{ij}]$$	
3.19	$$\tau_{ij} = \tau_{ji}$$	
3.20	$$\bar{\bar{\tau}} \cdot n = \lambda n$$	
3.21	$$\lvert \bar{\bar{\tau}} - \lambda \bar{\bar{I}} \rvert = 0$$	
3.22	$$F_s = \int_{\Sigma} \bar{\bar{\tau}} \cdot n\, d\sigma$$	

Eq. #	Mathematical Formulation	
3.23	$$\int_{\Sigma} \bar{\bar{\tau}} \cdot n \, d\sigma = \int_{V} (\nabla \cdot \bar{\bar{\tau}}) \, dV$$	
3.24	$$\rho dV \frac{Dv}{Dt} = (\nabla \cdot \bar{\bar{\tau}}) dV + \rho f_m dV$$	
3.25	$$\rho \frac{Dv}{Dt} = \nabla \cdot \bar{\bar{\tau}} + \rho f_m$$	
3.26	$$f_n = -pn$$	
3.27	$$\bar{\bar{\tau}} = -p\bar{\bar{I}}$$	
3.28	$$\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$$	
3.29	$$\tau'_{ij} = \alpha_{ijkl}\gamma_{kl}$$	
3.30	$$\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d + \lambda(\nabla \cdot v)\bar{\bar{I}}$$	
3.31	$$\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d + \left(\mu_B - \frac{2}{3}\mu\right)(\nabla \cdot v)\bar{\bar{I}}$$	
3.32	$$F = -\int_{\Sigma} pnd\sigma + \int_{\Sigma} \bar{\bar{\tau}}' \cdot nd\sigma$$	
3.33	$$M_{x_0} = -\int_{\Sigma} (x - x_0) \wedge (pn) d\sigma + \int_{\Sigma} (x - x_0) \wedge (\bar{\bar{\tau}}' \cdot n) d\sigma$$	
4.2	$$\frac{d}{dt}\left[\int_{V_f} \rho v dV\right] = -\int_{\Sigma_f} pnd\sigma + \int_{\Sigma_f} \bar{\bar{\tau}}' \cdot nd\sigma + \int_{V_f} \rho f_m dV$$	

Eq. #	Mathematical Formulation	
3.35	$$\frac{d}{dt}\left[\int_{V_c}\rho v dV\right]+\int_{\Sigma_c}\rho v[(v-v_c)\cdot n]d\sigma=-\int_{\Sigma_c}pnd\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot nd\sigma+\int_{V_c}\rho f_m dV$$	
3.36	$$\int_{V_0}\frac{\partial(\rho v)}{\partial t}dV+\int_{\Sigma_0}\rho v(v\cdot n)d\sigma=-\int_{\Sigma_0}pnd\sigma+\int_{\Sigma_0}\bar{\bar{\tau}}'\cdot nd\sigma+\int_{V_0}\rho f_m dV$$	
3.37	$$\frac{d}{dt}\left[\int_{V_f}\rho(x-x_0)\wedge v dV\right]=\sum M_{x_0,\text{ext}}$$	
3.38	$$\frac{d}{dt}\left[\int_{V_c}\rho[(x-x_0)\wedge v]dV\right]+\int_{\Sigma_c}\rho[(x-x_0)\wedge v][(v-v_c)\cdot n]d\sigma=\sum M_{x_0,\text{ext}}$$	
3.39	$$d\dot{Q}_{\text{cond}}=q_n(n,x,t)d\sigma$$	
3.40	$$q_n dA=q_1 dA_1+q_2 dA_2+q_3 dA_3$$	
3.41	$$q_n=q\cdot n$$	
3.42	$$\dot{Q}_{\text{cond}}=\int_{\Sigma}q\cdot nd\sigma$$	
3.43	$$\int_{\Sigma}q\cdot nd\sigma=\int_{V}(\nabla\cdot q)dV$$	
3.44	$$q=-k\nabla T$$	
3.45	$$\frac{d}{dt}\left[\int_{V_f}\rho(e+\frac{\lvert v\rvert^2}{2})dV\right]=-\int_{\Sigma_f}pv\cdot nd\sigma+\int_{\Sigma_f}v\cdot\bar{\bar{\tau}}'\cdot nd\sigma+\int_{V_f}\rho f_m\cdot vdV-\int_{\Sigma_f}q\cdot nd\sigma+\int_{V_f}(Q_c+Q_r)dV$$	

Eq. #	Mathematical Formulation
3.46	$$\frac{d}{dt}\left[\int_{V_c}\rho(e+\frac{\lvert v\rvert^2}{2})dV\right]+\int_{\Sigma_c}\rho(e+\frac{\lvert v\rvert^2}{2})[(v-v_c)\cdot n]d\sigma=-\int_{\Sigma_c}pv\cdot nd\sigma+\int_{\Sigma_c}v\cdot\bar{\bar{\tau}}'\cdot nd\sigma+\int_{V_c}\rho f_m\cdot vdV-\int_{\Sigma_c}q\cdot nd\sigma+\int_{V_c}(Q_c$$

Study notes written from the official course material (Notes.pdf, Chapter 3 · p. 34). Online version: subjects/fluid-mechanics/teoria/topic-3.html